

ISDRA Points Formula Review and Revision Proposal

July, 2002

Preamble

From reading the following article, one might conclude that we have never heard of the K.I.S.S. (Keep it Simple Stupid!) mantra. In fact, we are strong proponents of K.I.S.S.. However we also strongly believe in doing what is necessary to do a job right. Each element of the following discussion relates to aspects of the ISDRA Points system that are important determinants of the points assigned to races and ultimately to points earned by drivers. The original points formula was based upon purse, number of drivers and miles run. For the Unlimited Class, the 1999-2000 formula was based upon purse, number and quality of drivers and miles run. For the Limited classes, purse was left out of the formula for the past three years. This article examines the impact of those changes and makes recommendations for corrections where needed⁽⁰⁾.

We have heard from many members who support the new formulas even while being critical of some aspects. To that end, we examined the available data with an eye towards identifying what has worked well, what hasn't and what must be changed in order to assure a continued level playing field. And above all, keeping it as simple as possible!

Introduction

Gary Lillie of Minnesota designed the original speed category ISDRA point formula and J. Malcolm McDougall, Harris Dunlap and Dick Molburg formulated the medal program. The first ISDRA Medals were awarded in 1972. Only the Unlimited Class was awarded medals that season and George Attla took home the Gold, Dr. Roland Lombard, the Silver and W. Kent Allender received the Bronze. Since that time, a total of 616 ISDRA Medals have been awarded.

The formula served ISDRA well for many years and was based upon purse, the number of entries and the miles run for a race, all three being important variables in a sled dog competition.

While purses were small and roughly similar, this formula was appropriate. However, over the intervening decades of development within the sport, some of the characteristics of races changed, primarily in the area of purse. By 1999 a number of races had much greater purses than folks would have considered likely decades before and these purses had a dramatic effect on the points assigned to a race.

To mitigate the effects of the largest purses, under the original system, the point value for a race was capped at 350 points for the Unlimited class and 250 points for the Limited classes. However, there was no 'floor' for the races point values. As a result, some races were rated as little as 1/10 the value of the largest races. This resulted in a 'bimodal' distribution of race point values, which means that one group of races within a class had point values as low as 8 to 10 points and a second group of races had point values as high as 250 (limited classes) or 350 (unlimited classes) points.

This tremendous spread in race point values left some racers completely out of the hunt for medals unless they raced in every major point race. Please see the July 1999 issue of INFO, vol. 30, no.6, pp 22 –24 for the first discussion of the 1999 revisions with subsequent items in the October 1999

issue, p. 6 and the 2000-2001 Sanctioning Program, pp. 12-13. For the current discussion, keep in mind that a ratio of 3:1, i.e. the major races being worth 3 times the value of the minor races is perfectly adequate to separate the best drivers from the rest of the pack.

When the ISDRA point system was revised in 1999 for the speed category, the board directed that a thorough review of the new system be conducted when adequate data had accumulated under the new system. At the present time we have three seasons of data under the new system. Three seasons is sufficient, albeit minimally, to permit a basic analysis of the new point formula data and illuminate any strong trends which may be emerging.

But before we look at the extant data, let's review further the issues addressed in the revision of 1999. While the US/Canadian currency exchange rate was the impetus for a revision, the revision nonetheless attempted to address several points, the most significant being the inordinate spread between the lowest point value races and the highest. The primary culprit of the excessive spread was the reliance on purse, which tied in with the exchange rate issue.

It was proposed that a formula be based on the 'quality and quantity of competition'. In other words, the higher the quality and quantity of competition at a race, the higher the points associated with that race. Also, it was noted informally, that quality of competition was generally felt to be predictable by the size of a race purse, suggesting that a measure of driver quality could be a workable substitute for purse.

The ISDRA points system provided a ready-made ranking of 'driver quality'. Clearly, if we didn't consider our ISDRA rankings to be an indicator of quality, there would be no point in awarding Medals or certificates! So a formula using driver's net ISDRA points was constructed. The formula aggregated the net points earned by drivers in the previous season for the appropriate class. In the event a driver had less than 100 points, they were automatically counted as 100 points. By doing this, the formula acknowledged that while all drivers were not of equal value, any driver was worth at least 100 points.

To avoid huge numbers, scaling factors were used to reduce the aggregated point values. The first scaling factor also serves a 'benchmark' for races. As the entry size goes above the benchmark, the race will be worth somewhat more than a strict arithmetic average would produce and as the entry goes below the benchmark, the race will be worth somewhat less than what an arithmetic average would produce. The deviation from the 'benchmark' is a gradual linear function which we won't graph here, but should be kept in mind as it comes into play later in the present discussion. The second scaling factor was a simple multiplier (.66 or .75) designed to further reduce the size of the numbers, but not impart any further 'weighting'.

To the aggregated and scaled driver value, mileage was scaled up by a factor of 2 and added. Caps were placed on the maximum number of mileage points to discourage any attempts to tilt the overall points awarded to a race by grossly extending the mileage relative to the norm for the particular class.

Finally, a 'floor' support was added which assured that any race would be worth a minimum number of points just as any driver was considered to be worth a minimum number of points.

During the review period, Jim Welch of Alaska rightly pointed out that under any system, it was a foregone conclusion that in the Unlimited classes, the Anchorage World Championships and the Fairbanks Open North American Championships should come out on top. However, this proved to be

unattainable without reliance on purse and so for the Unlimited classes only, purse was incorporated, but to a lesser extent than under the old system. Additionally, US and Canadian dollars were treated equally, as it was pointed out that the significance of the dollar is relative to the country in which it is paid and that the 'economic strength' aspect of the two currencies was largely irrelevant in assigning a sporting value to competitions.

The 1999-2000 points formula approved by the board was as follows;

- a. **The net ISDRA Championship points from the ISDRA Points Standings for the appropriate race class are summed for all entrants with greater than 100 net points. In the event an entrant does not have standing from the previous season, they shall use their standing as of the Tuesday prior to the first full weekend of racing in February and use that standing through the remainder of the racing season.**
- b. **Drivers not meeting criterion a. shall each be counted as 100 points.**
- c. **The sum of a. and b. shall be divided by 20 (Unlimited Class) or 25 (Limited Class) and then multiplied by .66 (Unlimited Class) or .75 (Limited Class).**
- d. **The total miles run shall be multiplied by 2 and added to c., not to exceed 150 mileage points for the Unlimited Class or 75 mileage points for Limited Classes.**
- e. **For the Unlimited Class only, one point for each \$100.00 up to and including \$20,000.00 will be added to d., not to exceed 200 purse points.**
- f. **If the result in e. is less than 125 for an Unlimited Class, then the assigned points will be set to 125. If the result in d. is less than 75 for a Limited Class, then the assigned points will be set to 75.**

The revised formula was used for the first time during the 1999-2000 season. In the spring of 2000, the board approved an amendment adding a small amount of 'bonus' points to the top three finishers (+8%, +4%, +2%) to achieve a quasi-curvilinear distribution of points and more clearly reward the top finishers. However, this adjustment is applied in the distribution of points among finishers, not in the race points formula and will not be discussed further in this article.

To summarize so far, the new points formula was designed to avoid reliance on purse (at least for the Limited classes), reduce the gap between the largest and smallest races while not limiting the maximum value a race could be worth and affirm that all drivers, regardless of points earned previously, add a certain value to a competition.

As with anything, the new approach, while addressing the concerns of the time also introduced the potential for new problems. In this case, the most obvious was the potential for 'points inflation' or 'creep' in race values. Since there were no caps on maximum race point values, it was possible for races to continually escalate in value year after year. Furthermore, as a result of inflation in the point values for races, the maximum net points a driver achieved could also follow suit and increase year after year. Which then in turn would contribute to race point inflation and so on and so forth until we again have races worth a great number of points while other races, never attracting top point drivers, languish at the bottom.

Clearly, this could eventually bring the points system back to where it had started, i.e. with an excessive spread between the highest and lowest point value races.

Another concern was the effect that the lack of purse points would have on the actual purses paid by events. Since purse was no longer factored into the formula, the incentive to raise larger purses might be diminished. This was uncharted territory for the ISDRA points system.

A third concern was predictability. The lack of purse as a factor in the formula would make it more difficult to decide which races to go to if one was chasing points.

Review

Fast forwarding to 2002, we have three seasons of data under the conditions described above, so it is time to look at the data and see what happened.

Unlimited Class. First, let's look at the Unlimited Class point values for all races from the 1998-1999 season through 2001-2002. Figure 1 shows this data. For the '98-'99 season, under the original point formula, the highest point value was 350 points. Actually, there were two races valued at 350 points, the ONAC and the Rondy. The lowest point Unlimited Class race for the 98-99 season was the Lost Nation Forest Sled Dog race with 37.4 points. Note there were only 10 Unlimited class races that season, so the right-hand 'tail' of the graph goes to 0 after race #10.

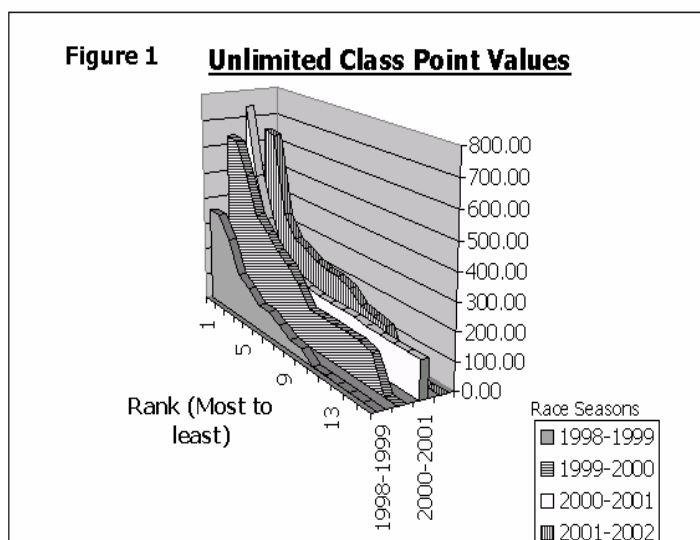
In the '99-'00 season with the revised formula, the top races (still the Rondy and the ONAC) jumped to point values of 631.24 and 624.47, respectively. But at the same time, the lowest point values ranged from 135.00 points to 118.80 points (reduced from 135.00 due to minimum trail requirements not met).

Moving to the '00-'01 season, the Rondy cancelled and the ONAC was the top point race at 737.85⁽²⁾. Again, the formula 'floor' came into play and supported 3 out of 16 races with the minimum value of 135.00 points.

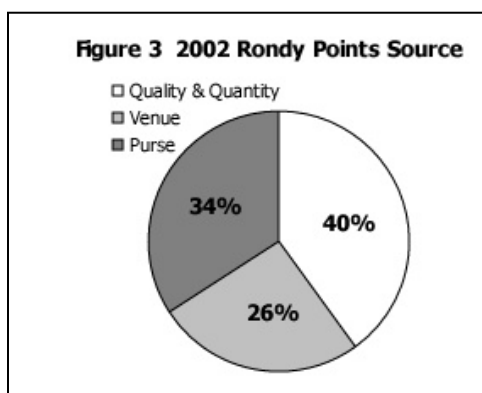
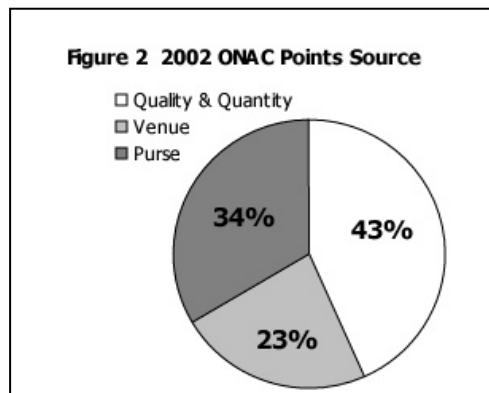
The Rondy was held during the '01-'02 season and once again, the Rondy and the ONAC held the top point values for the Unlimited class with 641.76 and 630.95 points respectively. The 135.00 point 'floor' supported 2 out of 12 races with one of those reduced further due to the minimum mileage penalty.

Of interest in Figure 1 is that although the maximum point value for unlimited class races jumped when the new formula was introduced, there is not any discernable point inflation occurring thus far. For the past three seasons, the point values for the Rondy and the ONAC have ranged from 624.47 to 737.85, peaked in '00-'01 and in '01-'02 were very comparable to their values in '99-'00.

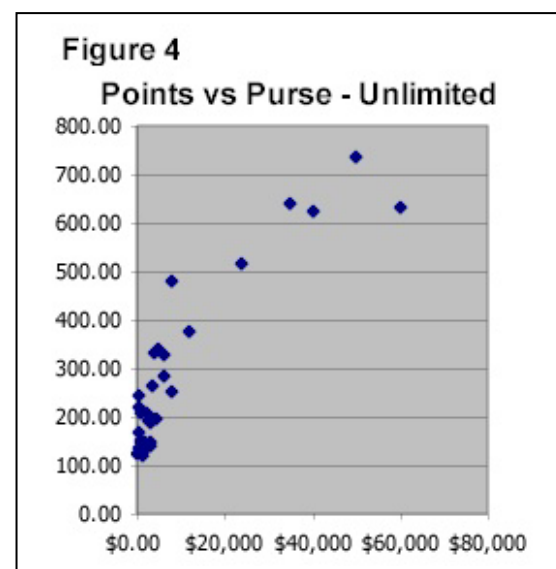
Also, it should be noted that the ratio between the largest and smallest point value races has been reduced from 9.35 under the original formula in '98-'99 to an average of 5.31 for the past three seasons for the Unlimited class.



Figures 2 & 3 show the Unlimited class points broken into three sources; a.) Driver quality and quantity, b.) Purse and 3.) Venue. By venue, we mean how far and how many times did teams run. Both Figures 2 & 3 show that the two events remain roughly comparable in all three areas. Purse contributed **34%** to the total points for each event, the drivers present contributed from 40% to 43% and the venue from 23% to 26% of the total point value.



Finally, let's look at the relationship of purse to points in the Unlimited Class. Figure 4 is a scatter plot of Points against Purse for all Unlimited Class races from the last three seasons. A clear relationship is present and the correlation coefficient for this data is $+0.9^{(3)}$.



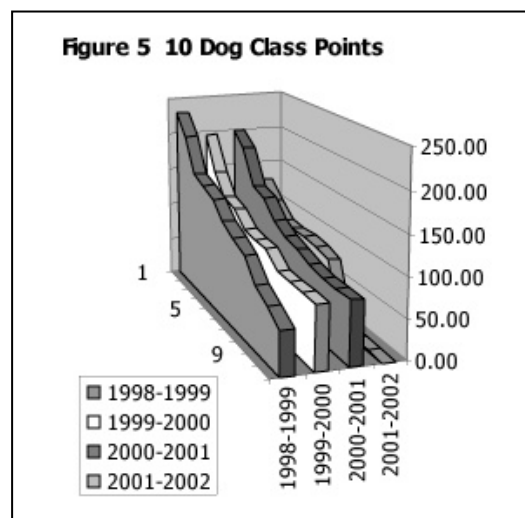
To summarize, for the Unlimited class the revised points formula has worked very well to date. The races that everyone can agree should be on top are on top, there has been no discernable inflation of point values over the first three years and purse is still a valid predictor of where one should race to pursue points as well as dollars. The ratio or spread between the top and bottom race point values has been reduced significantly, although it still exceeds the desired range of 3.0 to 4.0.

10 Dog Class. Figure 5 shows the Points for the 10 Dog Class from the 1998-1999 season through the 2001-2002 season. The introduction of the revised formula resulted in a slight decrease in the top value race, from 228.00 in '98-'99 to 195.45 in 99-00 and 198.14 in 00-01. In 01-02, the top race value dropped to 123.83 points, but this season was

also decimated by poor weather with 40% of sanctioned events cancelled. There is no point inflation evident in the data so far, in fact a case can be made that there may be some potential for point 'deflation', but the available data is not sufficient to force any conclusions.

The ratio of top to bottom point values has been; 4.13 for the '98-'99 season (original formula), 2.40 for the 99-00 season, 2.44 for '00-'01 and 1.52 for the decimated '01-'02 season.

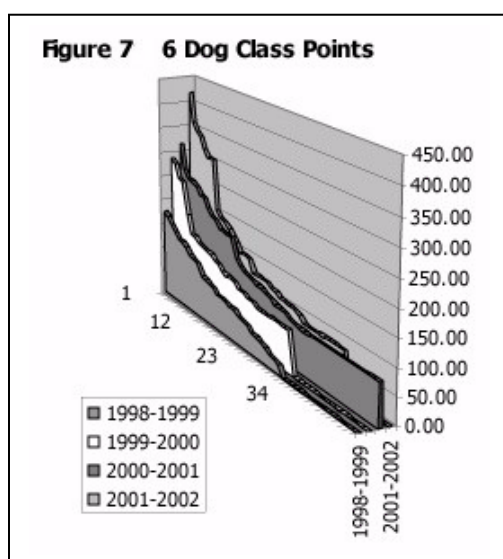
8 Dog Class. Figure 6a shows the same data for the 8 Dog Class. Again, there is no evidence of point inflation and some evidence suggesting slight point 'deflation'. In the 8 Dog class, we can see that a substantial number of races have been supported by the 'floor' values, 12 out of 24 (50%) in '99-'00, 14 out of 32 (43%) in '00-'01 and 4 out of 16 races (25%) in 01-02.



The ratio of top to bottom point values has been; 9.92 for the '98-'99 season (original formula), 2.91 for the '99-'00 season, 2.74 for '00-'01 and 2.61 for the '01-'02 season. Clearly, the spread has moved away from the 10:1 ratio of the previous system and is closer to the 3:1 standard proposed in 1999. However, the ratio is continuing to fall slowly, indicating that the distribution of race point values is flattening.

Figure 6b shows the source of points for the 2002 Colorado Blue 8 Dog Class. As can be seen the 99-00 formula heavily weights the points assigned to a race to the drivers entered.

6 Dog Class. Figure 7 shows the same data for the 6 Doggers and shows us the first evidence of point inflation.



The value of the top races has rapidly increased over the past three seasons. In '99-'00, the highest valued race was 284.50 points, in '00-'01, 313.78 points and in '01-'02 the top value race rose to 417.20 points. At the same time, there are substantial numbers of races supported by the floor, i.e. 7 out of 29 (24%) in '99-'00, 14 out of 44 (31%) in '00-'01 and 9 out of 36

races (25%) in '01-'02.

4 Dog Class. The 4 Dog class, shown in Figure 8 reveals the same trends as found in the 6 Dog class. The top 4 Dog class point values have climbed steadily; 250.00 in '98-'99, 274.55 in '99-'00, 333.20 in '00-'01 and 425.78 in '01-'02. At the same time, substantial numbers of races have been supported by the floor, i.e. 25 out of 39 (64%) in '99-'00, 22 out of 43 (51%) in '00-'01 and 7 out of 28 (25%) in '01-'02.

Skijoring. Skijoring has been in a transition phase for the past five years. It has been growing rapidly in the number of sanctioned races and the class has been split into 3, 2 and 1 Dog classes as demand increased and the number of individuals competing has grown. Nonetheless, the distribution of race point values has been very flat for the past five seasons and race point values have largely remained at the floor levels. This has resulted in more frequent ties in the final point standings since everyone is competing in races with identical race point values. To illustrate this, we have

Figure 6a 8 Dog Class Points

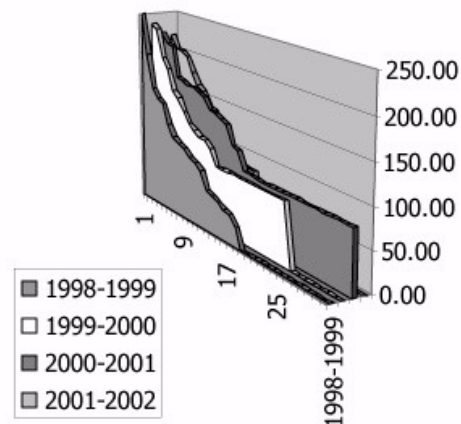


Figure 6b Colorado Blue IdidaRace 2002

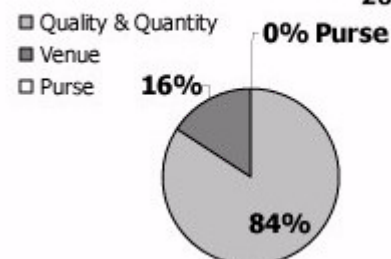
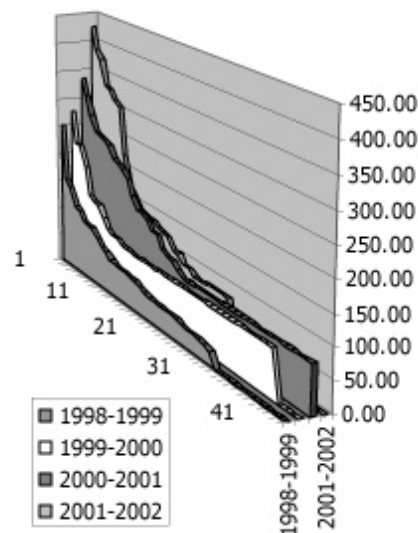
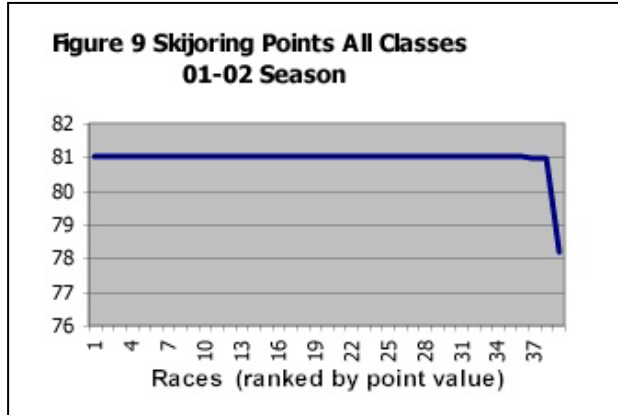


Figure 8 4 Dog Class Points



aggregated the 3, 2 and 1 Dog skijoring race point data for the '01-'02 season in Figure 9.



In Figure 9, the highest Race point value was 81.00 ⁽⁴⁾ for 38 out of 39 races (97%). It appears that 1 races had less than 81.00 points, but in actuality, it was also worth 81.00 points, but was adjusted to 78.20 points due to a tie for 1st place. When a tie occurs, the points are added for the entrants and the number of teams tied divides the total.

Obviously, there has been no point inflation or deflation in Skijoring and there is negligible spread in the race point values

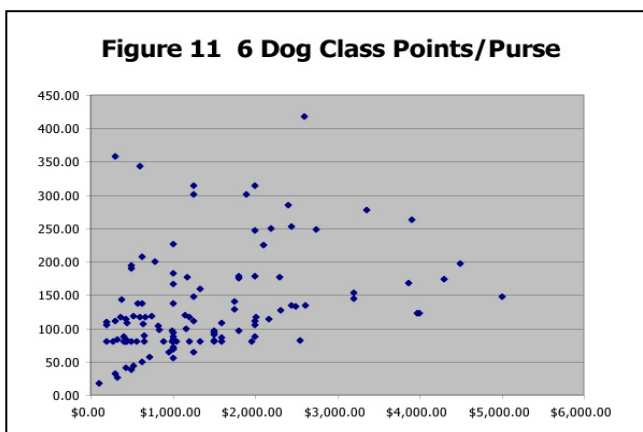
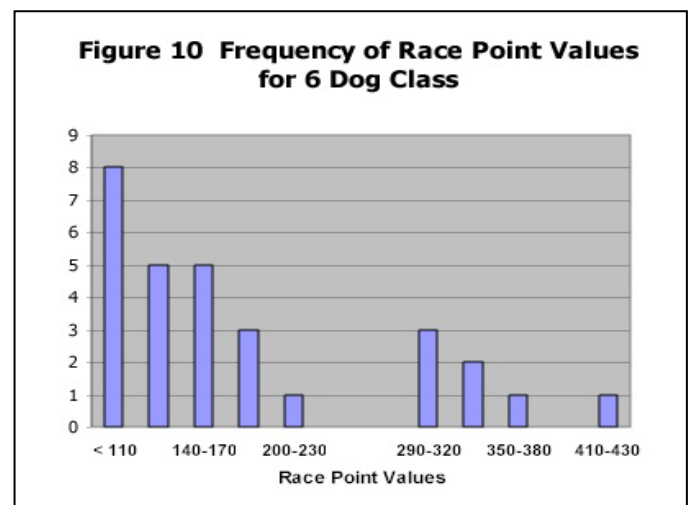
Summary

What does all this mean? It depends on which class interests you. The formula for the Unlimited Class appears to be working very well so far; no points inflation, the ratio between the top and the bottom has been reduced substantially and the ranking of the races is consistent with generally accepted expectations.

For the 10 Dog and 8 Dog classes, there is little to complain about. There is evidence of slight point deflation and for the '01-'02 season the spread of point values has dropped somewhat below the standard, but the small sample of races for the '01-'02 season is a likely factor.

For the 6 Dog and 4 Dog classes, there is definite evidence of substantial point inflation and the spread of race point values has been dramatically increasing because the benchmark for class size is too low.

For example, Figure 10 shows the distribution of '01-'02 6 Dog class point values broken into 30 point intervals. Even with a sample size of 29 races, we can see that the class appears to be breaking into two groups with a gap between. The larger group of races has point values less than 230 points, while a second, smaller group has point values larger than 290.



In general, drivers have continued to express dissatisfaction with the difficulty in predicting where they should race for maximum points. Although isdra.org provides a tool for estimating the point values for any given race, its use has been minimal presumably due to the time required to enter the necessary data. Understandably, drivers want to

know at a glance which race is likely to be a big point race and purse has always been the key.

Figure 11 illustrates this problem in a scatter plot of the purse vs points for the 6 Dog Class for the past three seasons. The correlation coefficient for this data is .38 and it's pretty obvious that there is little relationship between purse and points.

For Skijoring, the situation is different. The benchmark for class size is too high. As a result, all the races end up with their point values at the floor and the likelihood of ties in the standings is greatly increased.

Proposed Changes

Use the current Unlimited Class points formula for all classes with adjustments for the class size benchmarks and purse caps. Details follow, see Table 1 for summary of the constants to be used (**Note: these may be adjusted after we 'run the numbers' with race data from prior seasons**).

Unlimited Class. No changes.

10 Dog Class. Update class size benchmark to 20, leave mileage cap unchanged, set purse cap at 120 points.

8 Dog Class. Update class size benchmark to 20, set mileage cap at 60 points, set purse cap at 100 points.

Table 1		Adult Division		Benchmark	Mileage Points Cap	Purse Points Cap	Purse Cap
Type	Category	Class	Benchmark				
Sled	Speed	Unlimited	20	150	200	\$20,000.00	
Sled	Speed	10 Dog	20	75	120	\$12,000.00	
Sled	Speed	8 Dog	20	60	100	\$10,000.00	
Sled	Speed	6 Dog	40	50	80	\$8,000.00	
Sled	Speed	4 Dog	40	35	60	\$6,000.00	
Sled	Speed	3 Dog	10	30	40	\$4,000.00	
Skijoring	Speed	3 Dog	10	30	10	\$1,000.00	
Skijoring	Speed	2 Dog	10	30	10	\$1,000.00	
Skijoring	Speed	1 Dog	5	30	10	\$1,000.00	

6 Dog Class. Update class size benchmark to 40, set mileage cap at 50 points, set purse cap at 80 points.

4 Dog Class. Update class size benchmark to 40, set mileage cap at 35 points, set purse cap at 60 points.

3 Dog Class. Update class size benchmark to 10, set mileage cap at 30 points, set purse cap at 10 points.

3 & 2 Dog Skijoring. Update class size benchmark to 10, set mileage cap at 30 points, set purse cap at 10 points.

1 Dog Skijoring. Update class size benchmark to 5, set mileage cap at 30 points, set purse cap at 10 points.

Notes:

⁰ If you want to skip all that follows, here's the bottom line(s). The new points formula has worked very well for the Unlimited Class and not bad for the 10 Dog and 8 Dog Classes. The 6 Dog and 4 Dog Classes are showing 'point inflation', meaning some races are becoming worth more and more points each season. For Skijoring, practically all races are worth the same amount, which leads to more ties in the standings.

The formula is not broke, but needs to be 'tweaked'. The best thing is to emulate the Unlimited Class formula by reintroducing purse in a limited fashion and adjust the scaling constants where needed. The 6 Dog and 4 Dog scaling constants need to be increased to bring down the point inflation and the Skijoring scaling constants need to be decreased to bring race point values above the floor supports.

¹ Before constructing the 1999 ISDRA points formula a web search and review was undertaken of various points systems in use by other sporting organizations. The results were interesting in that there was no consistency in the approaches taken and no 'common thread' ran through the formulas reviewed. Attempts to find any general discussions on mathematical models for ranking and evaluating aggregated sports performances yielded nothing – perhaps interesting territory for a graduate research project.

² All the actual point values corresponding to these graphs can be found at <http://www.isdra.org/Events/Results/RacePoints.asp>. Select the season and the class descriptors to retrieve all event names, location, entry N, purse, total miles and race point values.

³ A correlation coefficient is an expression between -1.0 and $+1.0$ that indicates to degree to which two variables are related. A coefficient of 0 indicates no relationship; a coefficient of $+1$ indicates a perfect positive relationship.

⁴ The eagle-eyed reader will recall that the minimum points value in the current formula is 75. To that must be added the 8%, or 6.0 points for the first place finisher to arrive at the final minimum value for a limited class race (81.0 points).